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Perspective

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Immune control of brain physiology

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Abstract

The peripheral immune system communicates with the brain through complex anatomical routes involving the skull, the brain borders, circumventricular organs and peripheral nerves. These immune–brain communication pathways were classically considered to be dormant under physiological conditions and active only in cases of infection or damage. Yet, peripheral immune cells and signals are key in brain development, function and maintenance. In this Perspective, we propose an alternative framework for understanding the mechanisms of immune–brain communication. During brain development and in homeostasis, these anatomical structures allow selected elements of the peripheral immune system to affect the brain directly or indirectly, within physiological limits. By contrast, in ageing and pathological settings, detrimental peripheral immune signals hijack the existing communication routes or alter their structure. We discuss why a diversity of communication channels is needed and how they work in relation to one another to maintain homeostasis of the brain.

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Introduction

We used to think that the only role of the immune system was to protect the body from pathogens, cancer and damage. It is now clear that the immune system also functions to control tissue development, normal activity and maintenance. Indeed, immune cells and signals regulate crucial events in central nervous system (CNS) development, complex animal behaviours and brain ageing. Yet, such cells and signals have limited access to the brain tissues. There is a seeming incompatibility in the fact that the activities of the immune system often entail local tissue damage and that the function of the CNS is entirely encoded in its delicate structure with limited regenerative capacity. To account for this discordance, the CNS evolved a diverse and unique set of anatomical solutions that allow for physiological immune-to-brain communication with minimal risk of immune-mediated tissue damage.

In this Perspective article, we outline the recently determined anatomy of structures where these interactions occur, describe the mechanisms that use these communication routes to affect brain development, homeostasis and ageing, and suggest why such a diversity of anatomical routes might be necessary. We propose that immune–brain communication at these interaction points is always active, and the physiological or pathological influence of the immune system on the brain is dictated by the type of signal that is present at these structures or the change in the properties of the structures themselves. We envision that in the future, therapeutic agents will be designed to reach the brain along these entry points, or even heal the CNS from afar, by normalising the immune signals or properties of the routes of immune–brain crosstalk.

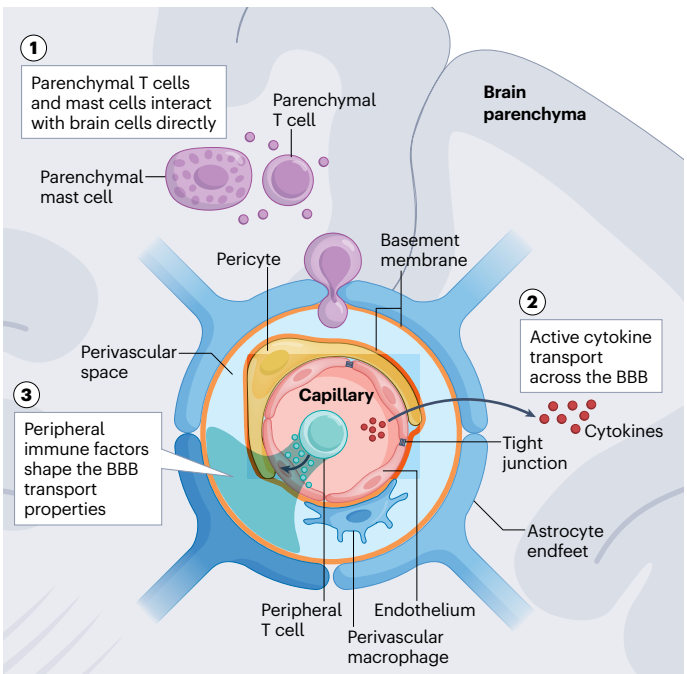


Fig. 1 | Blood–brain barrier as a site of direct blood–parenchyma communication. The multilayer structure of the blood–brain barrier (BBB) includes the endothelial and pericyte layer, the astrocytic endfeet (glia limitans) and the perivascular space between them. (1) Parenchymal immune cells may act as local 'glands', directly affecting brain cells around them. The current view is that these cells enter the parenchyma via the BBB. (2) The layers of the BBB may actively transport cytokines. (3) Peripheral immune factors affect the permeability of the BBB – for example, by blocking access of certain immune molecules and cells, while allowing the passage of others.

Structures mediating physiological immune–brain communication

Peripheral immune cells and signals shape brain physiology through specialized and highly selective anatomical routes. Some of these immune signals are present in the cerebrospinal fluid (CSF), which bathes the brain and fills the meninges. The meninges contain diverse immune cell populations, some of which originate directly from the skull marrow. Cytokines and immune cells are thought to enter the brain by crossing the blood–brain barrier (BBB). Peripheral factors can also affect the brain indirectly by modifying the features of brain structures. For example, immune signals may alter the barrier properties of the BBB and the choroid plexus (CP), which in turn change the cargo that is transported across, and the secretory properties of the CP epithelium, which is a vital producer of the CSF. Stimulation of peripheral nerves by immune cells and signals is another example of peripheral signalling that may change brain activity without entering the brain tissue. In the circumventricular organs (CVOs), immune stimuli are sensed by the brain cells, which may produce secondary molecules to propagate the signal further into the deeper brain structures, thereby spanning the direct and indirect modes of immune–brain interaction.

Brain-resident immune cells can shape brain activity directly

The CNS parenchyma is tiled by microglia, which are yolk sac-derived resident phagocytes that infiltrate the CNS during early embryonic development. Microglia have vital roles in brain development and maintenance and are highly attuned to the signals in their environment, which makes them the first responders to danger and initiators of immune responses. In this Perspective article, microglia are considered as final executors, acting according to instructions imposed by the peripheral immune system.

Besides microglia, T cells and mast cells are also present in healthy brains (Fig. 1). Resident effector memory ($CD69^{+}CD44^{hi}CD62^{low}$) T cells have been detected near blood vessels in the brains of both mice and humans[1–3]. A subset of these cells produce CC-chemokine ligand 5 (CCL5), and its receptor CC-chemokine receptor 5 (CCR5) is expressed by healthy neurons and microglia. CCR5 stimulation on neurons has been demonstrated to be essential for the temporal separation of memories[4]. Some brain-resident T cells are $FOXP3^{+}CD25^{+}$ regulatory T (T_{reg}) cells, which are classically recognized to attenuate inflammation by depleting interleukin-2 (IL-2; a survival factor for other T cells) and producing IL-10, amphiregulin and brain-derived neurotrophic factor, factors that are sensed by neurons and promote their function. In the context of stroke, multiple sclerosis and traumatic brain injury, overexpression of IL-2 in the brain to boost T_{reg} cells protected mice from inflammation-related pathology[1,2], and stimulation of receptors for IL-10, amphiregulin and brain-derived neurotrophic factor promoted neuron survival[1].

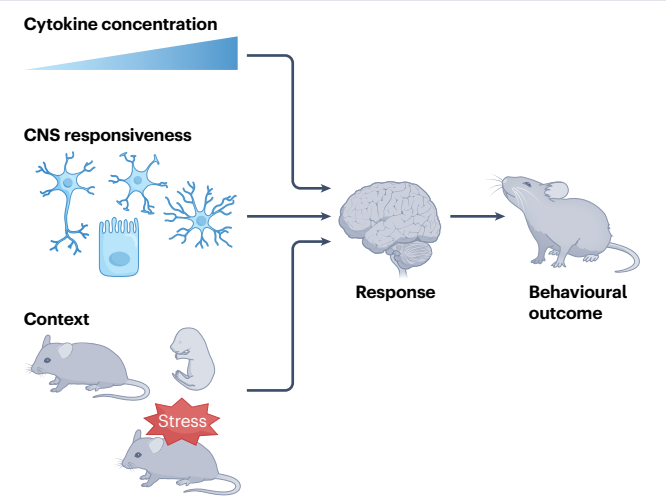
Mast cells can be detected in thalamic and hippocampal brain areas, and their numbers, locations and activity have been proposed to vary across species, endocrine status and stress levels[5]. Behavioural analysis of the $Kit^{W\text{-}sh/W\text{-}sh}$ mouse model, in which mast cells are deleted, revealed increased anxiety levels in these animals. This effect was phenocopied in wild type mice that were injected intracerebroventricularly with an agent that prevents mast cell degranulation but not in mice that were injected systemically[5]. Although the exact mechanisms by

Box 1 | Cytokines in the central nervous system

The finding that cytokine receptor expression is heterogeneous across central nervous system (CNS) cell types and regions suggests that the receptors have specific homeostatic roles. Yet, how cytokines reach specific locations within the CNS parenchyma and the mechanisms through which they exert their diverse functions are mainly unexplored.

One can speculate that the way in which cytokines shape neuronal activity probably depends on the concentration of the cytokine, the responsiveness of various CNS cell types to that cytokine and the overall context of its activity (see the figure). Considering concentration, each cytokine may have a local working range of 'beneficial' homeostatic activity. For example, IL-4 was reported to be present at low levels in the mouse cerebellum during development; an increase in its concentration resulted in defective synaptic pruning by microglia and the development of an attention deficit hyperactivity disorder-like phenotype. By contrast, deletion of IL-4 receptor on microglia did not result in behavioural abnormalities in this study[96], which suggests that IL-4 is only tolerated by the developing cerebellar microglia within a certain range, beyond which its function becomes detrimental. Similarly, within the choroid plexus (CP), an optimal range of IL-4 signalling is required to induce the production of brain-derived neurotrophic factor, a neurotrophic factor associated with learning. Both excess and absence of IL-4 were shown to halt the production of this factor by the CP epithelium[41].

Notably, cytokines released from their source form a gradient over a certain physical distance; parenchymal T cells and mast cells could serve as local glands that supply crucial factors directly to neighbouring CNS cells, whereas border-associated cells may deliver signals through a gradient that may act on neighbouring areas, such as the subventricular zone for the CP[137] or the cortical layers for the meninges[62]. Furthermore, the responsiveness of neural tissue to immune signals will also be conditioned by the density of receptor expression, and the presence and properties of the intracellular machinery that translates the signal into a response. For example, microglial responsiveness to TNF was reported to be negatively regulated by the transcription factor MEF2C. In ageing, the activity of MEF2C seems to be diminished, which renders microglia more sensitive to TNF stimulation[127]. Considering these different parameters that determine cytokine responses in the CNS, it is unsurprising that the same cytokine can have widely different effects in different contexts. For instance, interferon-γ promotes sociability by acting directly on GABAergic neurons in the prefrontal cortex when it is produced by meningeal T cells[64] but it was shown to promote synaptic weakening and withdrawal symptoms when it is produced by fragile-like regulatory T cells directly in the nucleus accumbens in an addiction model[138].



We are just beginning to understand the role of cytokines in the CNS and many questions remain unanswered. How do cytokines reach their receptors in the CNS, relative to the flow of cerebrospinal fluid? What is the distribution and nature of potential cytokine transporters across border tissues and what determines their activity? What are the physical activity ranges for the cytokines that are expressed by parenchymal immune cells versus border-associated cells, and how do they differ across locations and the lifespan? The use of innovative tools of cell labelling, imaging and spatial omics may help us to answer these questions in the future.

which mast cells could shape behaviour are elusive (Box 1), they are known to produce neurotransmitters such as serotonin, dopamine and histamine, as well as neuroactive peptides[6].

The blood–brain barrier: a selective border structure

The BBB is a specialized structure that is formed at the vasculature of the CNS. It is composed of tightly connected endothelial cells, which together with pericytes, produce a common basement membrane at the abluminal side of the endothelial layer that restricts the passage of cells and signals coming from the blood. Astrocytic endfeet form the glia limitans with their basement membrane, creating a compartment between the two basal membranes called the perivascular space, which is populated by myeloid cells and lymphocytes[7]. Although once considered to be a static, inert border, the current view is that the barrier and transport properties of the BBB are dynamic and adjust to the needs of the brain and immune signals in the periphery (Fig. 1).

Several studies have shown that the BBB can selectively transport cytokines and chemokines, including IL-1α, IL-1β, IL-1 receptor (IL-1R) antagonist, tumour necrosis factor (TNF), IL-6 and CCL2, as well as CCL5 (linked to physiological memory regulation) and CCL11 (associated with ageing and neurological symptoms of long COVID)[8–10]. In addition, cytokines can shape BBB function and recent single-cell RNA-sequencing analysis of human endothelial cells demonstrated the expression of IL-1R1 and IL-6R in venules, veins and large veins of the BBB[11]. Remarkably, endothelial cell-specific conditional deletion of *Il1r1* ameliorated symptoms in a mouse model of epilepsy[12], which suggests a physiological role of IL-1β in neuronal excitability via endothelial IL-1R1. In addition, IL-1β could promote pericyte-specific release of several immune factors such as CCL2 (refs. 13–16) that are essential for neuronal excitability guidance and synaptic transmission, pain and anxiety[16–18]. A local source of IL-1β in physiological conditions is macrophages, which have been reported to secrete this

Box 2 | Cerebrospinal fluid flow

Recent advances in our understanding of the border anatomy of the central nervous system (CNS) has led to an updated, albeit incomplete, model of cerebrospinal fluid (CSF) flow (reviewed elsewhere[96]). The CSF is generally thought to be largely produced by the choroid plexus[137,139]. The CSF flows through the ventricular system and fills the subarachnoid space to enter structures known as Virchow–Robin spaces, which are extensions of the subarachnoid space and surround the arteries that enter the brain parenchyma. From there, the CSF moves through the CNS tissue as interstitial fluid, creating a convective flow towards perivenular spaces through the glymphatic system[140]. Of note, the rate of arterial wall movement and CSF flow patterns are closely aligned, which suggests that arterial pulsation is the primary driver of CSF flow[141,142], although another recent study found that neuronal activity can increase interstitial fluid perfusion locally[143].

Originally, the CSF was thought to exit the perivenular spaces to the subarachnoid space and then directly to the venous sinuses in the dura mater through structures called arachnoid granulations[144], yet recent tracer imaging studies have revealed that the CSF reaches the dura mater first[50,70]. In mice and humans, parenchymal veins connect to dural sinuses through bridging veins, which cross the subarachnoid space and the arachnoid barrier. As the bridging veins pierce the arachnoid, they create discontinuities called arachnoid cuff exit (ACE) points, where CSF from the subarachnoid space can enter the dura mater in the vicinity of dural lymphatics, which run along the dural sinuses[70]. ACE points were found to be rich in immune cells, which from there can monitor the composition of the CSF, an indicator of CNS health[48–50]. Interestingly, the bridging veins were shown to keep a vascular structure typical for the CNS until the ACE point, whereas, within the dura mater, these vessels resembled those that are found outside of the CNS[70]. The role of arachnoid granulations, which are not found in mice, remains to be studied.

Imaging of tracers injected into the CSF and blood of mice and humans over time revealed that the ACE points enable bidirectional molecular transport[50,70]. On the one hand, CSF particles, including immune signals, were shown to flow via the ACE point to the dura mater on their way to the meningeal lymphatic vessels and, later, to cervical lymph nodes[46,70,145–147]. In addition, studies focusing on the skull bone marrow response to CNS infection and injury have shown that a portion of the CSF can reach the marrow and instruct local haematopoiesis[57,58], which was later confirmed in humans by intrathecal injection of contrast agent and magnetic resonance imaging[97]. On the other hand, cytokines produced by meningeal immune cells or injected transcranially were shown to regulate mouse behaviour[62–65] and shape the recovery process after cerebrovascular injury[148], which suggests that the transport of cytokines against CSF flow can also occur. Indeed, low quantities of tracers injected into the blood were observed in ACE points and on the surface of the brain, which suggests a passive transport from the dural blood vessels to the CNS[70]. Finally, it is thought that CSF can also exit the CNS along the cranial and spinal nerves, and a recent study detected CSF within the peripheral nerve structures[149].

cytokine in a glucose-dependent manner[19], and may be important for increased BBB permeability after food ingestion[20]. Interestingly, high-sugar diets have been shown to correlate with a higher risk of cognitive impairment[21], possibly by disrupting WNT–β-catenin signalling at the BBB via IL-1β[22]. IL-1β-induced signalling in endothelial cells has also been shown to be sufficient for the induction of sickness behaviour in mice[23], whereas congenital, systemic deletion of *Il1r1* resulted in altered memory processes as well as in disruption of short-term and long-term plasticity in the hippocampus[24]. Therefore, the effects of physiological cytokine signalling on brain activity may occur, at least in part, by modifying BBB functions. Beyond cytokines, a study has shown that a small subset of perivascular macrophages express high levels of scavenger proteins and are essential for the maintenance of CSF flow dynamics via the regulation of arterial motion and degradation of the extracellular matrix around the vessels[25].

In addition, changes in local blood supply driven by neuronal activity, known as neurovascular coupling, are regulated, at least in part, by nitric oxide[26], which can be produced by macrophages and microglia in response to IL-1β[27]. Therefore, under homeostatic conditions, IL-1β could increase local blood supply, drive microglia to form direct contacts with endothelial cells, and promote BBB permeability by disrupting endothelial tight junctions[28–30]. This pathway may support the brain's metabolic demands through immune-modulated changes in BBB properties. In the future, spatial 'omic' techniques may help to decipher the specific roles of immune cells and the signalling pathways induced in the regulation of BBB function.

The cerebrospinal fluid contains immune signals that regulate brain function

The brain ventricles, spinal cord and meninges are at all times bathed in CSF (Box 2). The CSF distributes nutrients, hormones and neurotrophic factors, such as cholesterol, thyroid hormones, insulin and insulin-like growth factor, while cleansing the tissue of metabolites, excess neurotransmitters and toxins[31]. Therefore, the composition and movement of CSF is critical to brain development and health. Immune cells and their products were reported to be present in the CSF under physiological conditions, and their composition changes throughout life[32] (Fig. 2). Analysis of the CSF of healthy individuals revealed predominantly central memory $CD4^+$ and $CD8^+$ T cells, natural killer cells, B cells, plasma cells, dendritic cells (DCs) and various monocyte populations[32,33]. A single-cell RNA-sequencing study has shown that at baseline some of these CSF subsets express cytokines that had previously been shown to regulate neurons, such as IL-15 (ref. 32). IL-15-mediated signalling was shown to facilitate hippocampal-dependent memory, and the effect was at least partially mediated by direct receptor activation on hippocampal neurons[34]. Furthermore, a subset of CSF monocytes expresses chemokines such as CCL3 and CCL4 (ref. 32), which may also have a role in memory formation via neuronal CCR5 stimulation[4]. CSF-dwelling cells in healthy humans also express cytokines such as IL-10, IL-16, IL-18 and IL-1β[32]. According to a transcriptomic dataset, microglia express receptors for IL-18 and IL-10, and IL-10Rα expression can also be found on a subpopulation of neurons[35]. The implications of this cytokine signalling at the steady state are unknown; they may act as tonic signals, which keep the system ready to respond quickly in case of

danger. However, the spatial distribution of receptor expression suggests that their activation might have a specific physiological function (Box 1). Other aspects of CSF composition and properties are, at least in part, controlled by the immune system indirectly via regulation of the brain borders, by changing the profile of molecules that pass to the CSF while selectively blocking the transfer of others, or by changing their secretory and detoxification activity.

The choroid plexus at the blood–cerebrospinal fluid barrier

Located within each of the four ventricles, the CP consists of fenestrated capillaries that are encapsulated by a monolayer of tightly connected and ciliated epithelia that form the blood–CSF barrier. The CP epithelium serves as the producer of CSF as it enables the transport of water, ions, nutrients and regulatory molecules via transporters and transcytosis while simultaneously producing hormones and growth factors that buffer and nourish the CNS via the CSF[36,37]. These activities are, at least in part, regulated by the peripheral immune cells and signals (Fig. 2).

The CP is populated by various immune cells, most of which are border-associated macrophages[7,38,39]. A subset of those, the Kolmer (epiplexus) cells, are of embryonic origin, reside on the apical (CSF-facing) surface of the epithelium and have a transcriptomic signature that is similar to disease-associated microglia, which suggests a high phagocytic activity under homeostatic conditions[7,38]. The specific functions of Kolmer cells remains to be studied. Still, as resident macrophages, they are likely to be important for tissue maintenance in the CP while remaining sensitive to immune stimuli, which can further regulate their activities.

By contrast, border-associated macrophages in the stroma line the vessels of the CP and sample blood-derived factors through the fenestrations[37]. Within hours following an intraperitoneal injection of lipopolysaccharide, which is a strong pro-inflammatory stimulus, these cells were shown to cover the CP vessels, potentially preventing access of the noxious stimuli to the CP epithelium[37]. Interestingly, in a mouse model of colitis, the CP vasculature initially opened but, within three days, closed through loss of the fenestrae protein plasmalemma vesicle-associated protein 1 (PV1)[36]. The initial opening of the CP vasculature may allow for stromal border-associated macrophages that line the vessels to sample antigens in the blood, whereas the subsequent tightening of the CP vasculature would prevent the potentially excessive effect

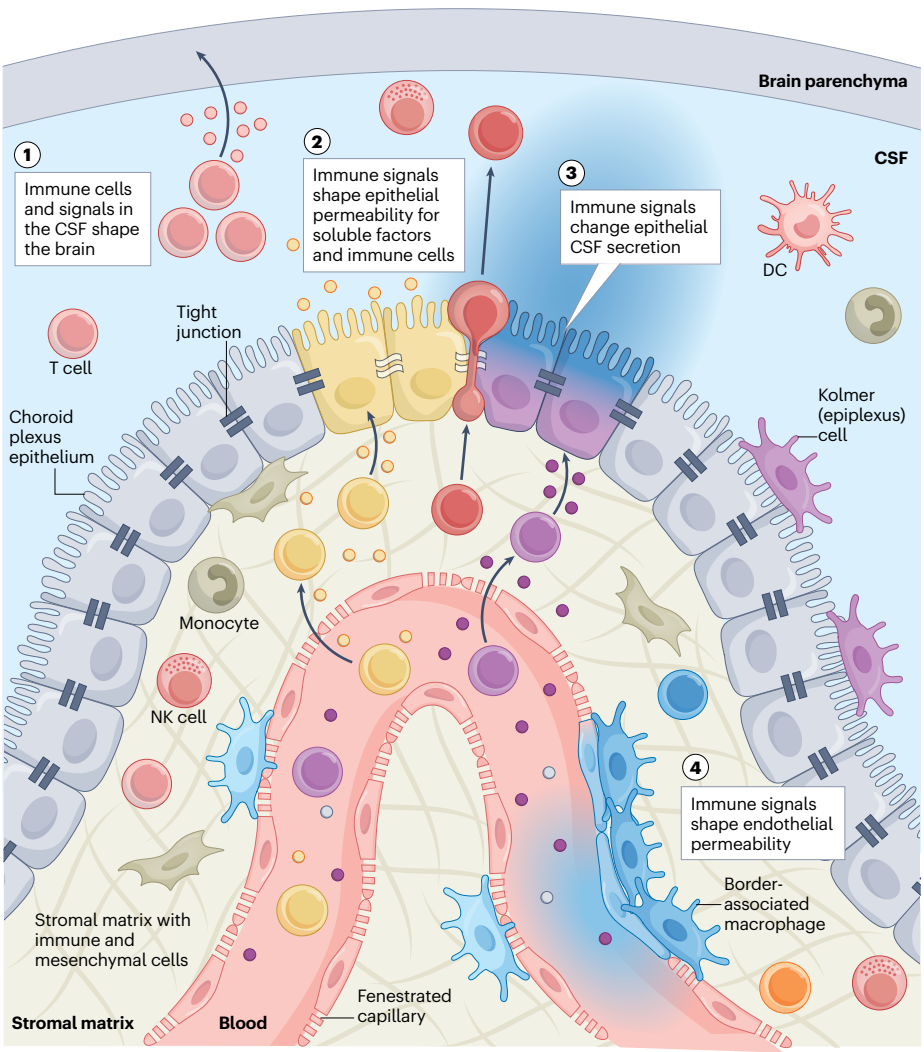


Fig. 2 | Peripheral immune signals at the choroid plexus may regulate the brain indirectly, by changing the composition of the cerebrospinal fluid. The choroid plexus (CP) comprises permeable fenestrated endothelial vessels that are embedded in a stromal matrix and covered by a monolayer of highly secretory epithelial cells, which create the blood–cerebrospinal fluid barrier. (1) Immune cells and signals that are present in healthy cerebrospinal fluid (CSF) can affect brain function. (2) Peripheral signals present in the circulation and stroma can modulate CP epithelium permeability and expression of trafficking molecules, which allow access of peripheral soluble factors and immune cells to the CSF. (3) Those signals may influence the secretory activity of epithelial cells and thereby contribute to changes in CSF properties, including its composition and flow. (4) Peripheral signals were also described to affect CP function by regulating the permeability of the fenestrated endothelium, both by recruiting macrophages to seal the vessels and by controlling the opening and closing of the fenestrations. DC, dendritic cell.

of inflammatory stimuli on the CP epithelium and the entry of pathogens into the brain. Mice with an endothelial cell-specific β-catenin gain-of-function, which is associated with loss of PV1 expression and development of a BBB-like phenotype in the CP vasculature, were shown to develop an anxiety-like behaviour[36]. This suggests that psychosocial disturbances that are associated with chronic inflammatory conditions (and CP enlargement in the case of depression in humans[40]) may be, at least in part, a result of the closure of CP fenestrations, which entails a reduced supply of nutrients to the CP causing the brain to starve.

Beyond macrophages, CNS-specific effector memory $CD4^+$ T cells were proposed to interact directly with the CP epithelium through interferon-γ (IFNγ)[41,42]. Knockout of *Ifng* or of its receptor (*Ifngr1*) in mice resulted in a loss of trafficking molecule expression by the CP and a decrease in the numbers of CSF-dwelling T cells and monocytes[42]. Trafficking of immune cells via other routes was not evaluated in these mouse models, however in another study, CP epithelium-specific inhibition of corticosteroid signalling following episodes of stress was shown to favour recruitment of T cells to the CSF and attenuate stress-induced behavioural deficits[43]. Other immune cells have also been shown to populate the adult CP under physiological conditions in both mice and humans, including natural killer cells, DCs and B cells[7,38,39], but their respective roles remain unexplored. The mechanisms underlying communication between immune cells and the CP epithelium are still elusive, yet the expression of cytokine receptors such as IL-1R1, IL-6R, IFNγR and CXCR2 by the CP epithelium[42,44] suggests that changes in the secretory and barrier properties of the epithelium may be partly due to physiological fluctuations in peripheral immune signalling. Another communication route may also involve immune–stromal interactions, as suggested by ligand–receptor analysis of single-cell RNA-sequencing data indicating that IL-1β signalling from macrophages to the endothelium occurs in the aged CP[39]. The full network of interactions between the immune cells, the epithelium and the stroma, and how these interactions shape CP physiology and CSF composition, remain understudied.

Skull bone marrow and meninges allow direct marrow–brain communication

Studies have revealed unexpected anatomical connections between the CSF, the meninges and skull bone marrow (Fig. 3). Bone marrow is found in long and flat bones, including the skull and vertebrae, where haematopoiesis occurs within a specialized stromal and endothelial niche. Compared with other bones, the skull marrow is distinct in terms of its haematopoietic output and, perhaps counterintuitively, skull-derived immune cells show higher expression of genes related to migration and inflammation[45].

The meninges are connective tissue layers that are located between the skull and the brain. The outermost layer is the dura mater, which contacts the bone, whereas the arachnoid and pia mater (together known as leptomeninges) cover the brain and spinal cord. The dura mater comprises the dural sinuses, which drain the blood from cerebral veins and CSF-draining lymphatic vessels that run along them[46,47]. CNS antigens first access the dura via the CSF, enabling an initial sampling by resident immune cells[38,48]. From there, these antigens can be drained to the cervical lymph nodes through the lymphatic network[49–51] (Box 2). Dura mater immune cells comprise a diverse population of granulocytes, lymphoid and myeloid cells (including antigen-presenting cells)[48]. By contrast, the inner leptomeningeal layers are less diverse and dominated by myeloid cells[38]. In addition, the skull marrow is directly connected with the meninges in both mice[52–55] and humans[45] by ossified vascular channels. Parabiosis, selective irradiation and skull bone transplantation experiments have revealed that in healthy mice, a substantial pool of monocytes, neutrophils and B cells, and to a lesser extent, macrophages and DCs, found in the dura mater are skull-derived[54–56]. Moreover, CNS-derived proteins may reach the skull marrow niche through the CSF to regulate haematopoiesis, and in turn, meningeal immune cell composition and brain function[57,58]. However, the cues that promote cell recruitment from the skull marrow rather than the blood are still unknown.

Meningeal mucosal-associated invariant T cells have been found to express antioxidant machinery, and thereby preserve meningeal barrier function under physiological conditions. Mice deficient in these cells showed an accumulation of reactive oxygen species in the meninges, reduced expression of junctional proteins, meningeal barrier leakage and, consequently, microglia activation and loss of cognitive ability[59]. Furthermore, increasing evidence indicates that meningeal-derived factors such as cytokines, chemokines and growth factors drive key events in brain development[60,61], contribute to neuronal connectivity and shape complex animal behaviours[62–67] by directly acting on the parenchymal cells from the CSF. For example, in a series of experiments in mouse models, loss of meningeal IFNγ was associated with loss of sociability in otherwise healthy mice, a link that was also observed in rats, zebrafish and flies. The authors proposed a fascinating evolutionary explanation to these observations: the presence of IFNγ-dependent antiviral immunity could allow for social behaviour, as without antiviral systems, physical closeness would create a risk of infection spread[64]. Other studies that combined local and systemic depletion strategies proposed that meningeal IFNγ and γδ T cell-derived IL-17 were also implicated in the regulation of anxiety, and IL-17 and IL-4 in promoting cognitive ability[62,63,65,66]. IFNγ, IL-17, IL-4 and IL-13 can be directly sensed by neurons[62,64,65,68], with IL-13 acting as a synaptic protein[69]. Although meninges are typically viewed as an exit point for the CSF, the movement of meningeal-derived cytokines into the CSF and the brain parenchyma, against the CSF outflow was recently proposed to take place at subarachnoid space–dura mater connections called arachnoid cuff exit points[70] (Box 2).

Circumventricular organs enable direct and indirect periphery–brain communication

The CVOs are specialized structures in the parenchyma with fenestrated vessels, which allow direct interaction of immune factors in the blood with the CNS. The subfornical organ, organum vasculosum laminae terminalis and area postrema are described as sensory CVOs that detect signals from the periphery. In contrast, the pineal gland, median eminence, neurohypophysis and subcommissural organ are considered to be secretory CVOs, which release regulatory molecules into the bloodstream. Except for the pineal gland and neurohypophysis, the CVOs are lined by astrocytes and a subset of specialized ependymal cells called tanycytes, which act as a barrier and a mediator between the blood, the CSF and the brain parenchyma[71] (Fig. 4).

Direct evidence of immune communication with CVOs has been provided mainly in the setting of systemic inflammation, when CVOs sense microbial molecules via Toll-like receptor 2 (TLR2) and TLR4, as well as cytokines via TNF, IL-1β and leukaemia inhibitory factor receptors, which results in fever induction, hypothalamic–pituitary–adrenal axis activation and development of sickness behaviour[72–74]. In the mouse median eminence, it was found that several cytokine receptor-encoding genes such as *Il11ra1* and *Ifngr1* were expressed at baseline and upregulated with age[75].

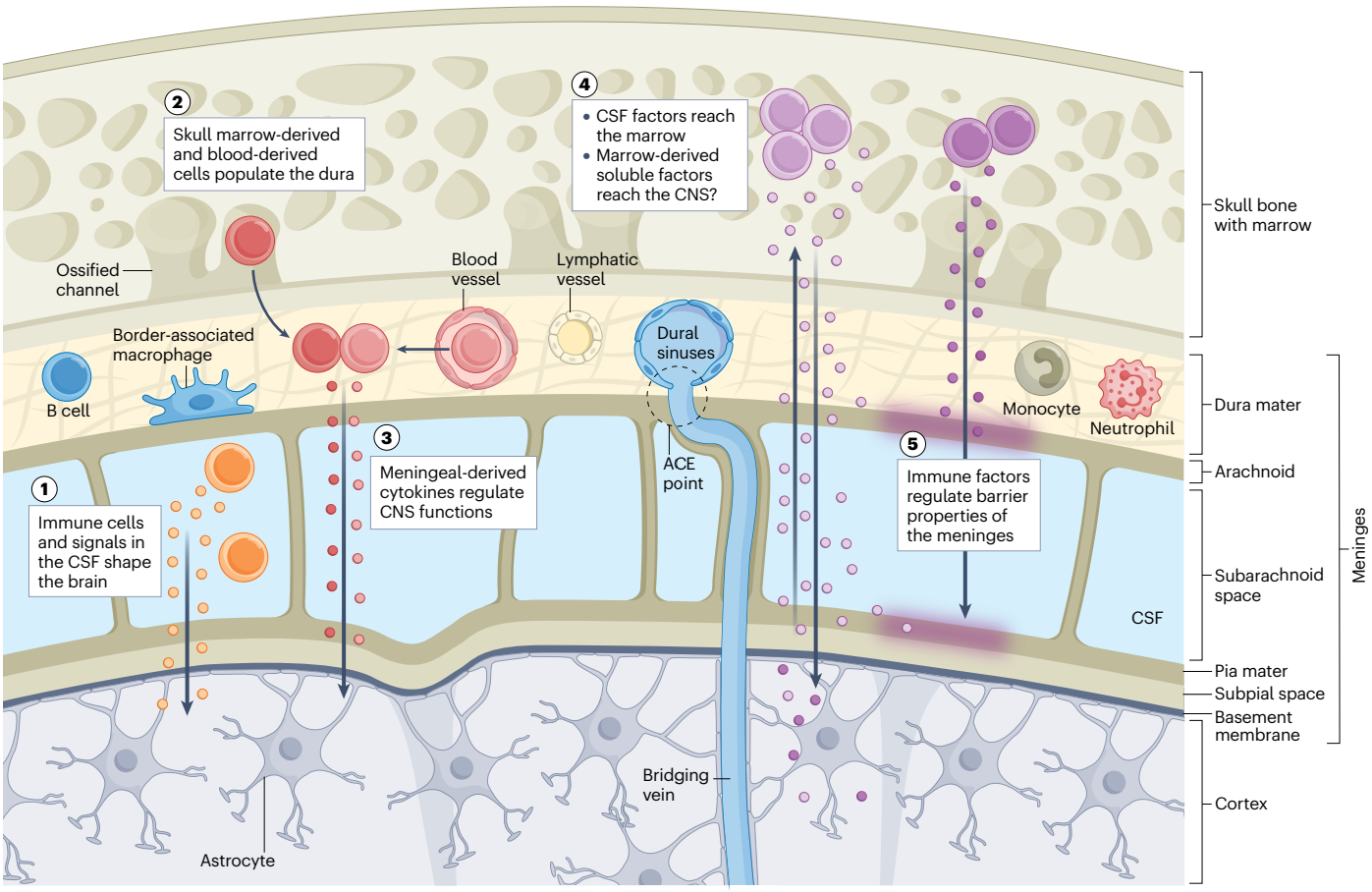


Fig. 3 | Meningeal anatomy allows bidirectional communication along the skull marrow–dura mater–cerebrospinal fluid–brain axis. Bone marrow is present in the skull bone, which is separated from the brain by three meningeal layers: the dura mater, the arachnoid and the pia mater. The space between the pia mater and the dura mater, which is known as the subarachnoid space, is filled with cerebrospinal fluid (CSF). (1) The immune cells and their products present in the CSF may shape brain function. (2 and 3), The dura mater is rich in immune cells, which infiltrate from both the skull marrow and the blood. Cytokines that are derived from dura mater-dwelling cells were reported to affect animal behaviour. (4) According to recent imaging studies, leakage of CSF into the dura mater and then into the skull marrow niche is enabled at arachnoid exit cuff (ACE) points, which are defined as discontinuities in the arachnoid barrier that are found around bridging veins as they pierce the arachnoid to enter the dura mater. Transport of cells and molecules was also proposed to occur in the opposite direction, from the skull and dura mater to the CSF. (5) Local immune cells and signals can also affect meningeal barrier properties, changing their permeability. CNS, central nervous system.

In mouse CVOs, microglia have a rounded morphology with processes that are shorter than in other brain regions and show more robust expression of markers such as CD16, CD32, CD86, CD206 and Ym1 (ref. 76), which is consistent with the notion that the CVOs may interact with peripheral signals that impose this phenotype on microglia. In the area postrema, perivascular $CD163^+CD169^+$ macrophages were shown to restrict the permeability of dextran molecules that are larger than 10 kDa (ref. 77), which suggests that, similarly to the CP[37] and meninges[70], the passage of substances is well controlled, in part thanks to the activity of the resident macrophages. In addition, $CD8^+$ T cells are present in the healthy ventral medulla oblongata and area postrema[78] – and possibly in the median eminence, organum vasculosum laminae terminalis and subfornical organ[79] – but little is known about the role of these cells in the regulation of CVO functions. A role for T cells in the regulation of blood pressure was previously proposed by a study demonstrating that angiotensin II administration leads to T cell activation and mice without T cells, or with T cells in which the gene encoding type 1 angiotensin II receptor was deleted, are resistant to angiotensin II-mediated hypertension[80]. Later, the same group showed that lesions in the brain region that includes the organum vasculosum laminae terminalis and that receives inputs from the subfornical organ, two of the CVOs known to be involved in blood pressure regulation, prevented both angiotensin II-mediated T cell activation and hypertension[81]. This suggests an interplay between CVO activity and T cells in this model, as well as, possibly, in the modulation of blood pressure within the physiological range. Future research will determine whether and how peripheral immune cells and signals contribute to other aspects of CVO function, which in turn, regulate the physiology of the body.

Immune–brain communication via peripheral nerves

The peripheral nervous system (PNS) informs the CNS about movements, senses and unconscious processes, such as autonomic functions of the organs and systems in the body. In turn, the CNS receives, processes and responds to the PNS, governing the activity of the body

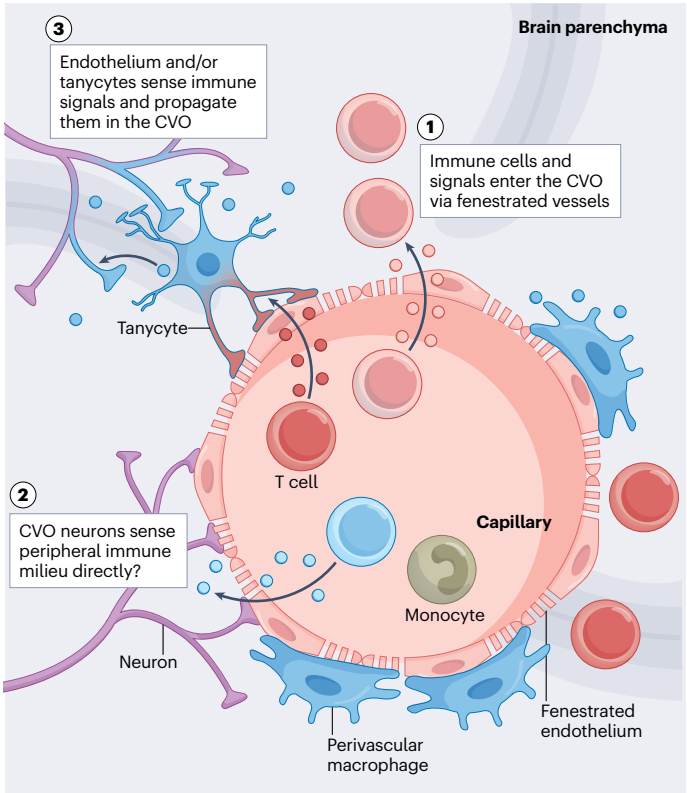


Fig. 4 | Immune–brain communication in circumventricular organs allows the brain to sense immune signals in the blood both directly and indirectly. (1) In circumventricular organs (CVOs), peripheral cells and signals can enter the brain parenchyma via fenestrated vessels. (2) It was proposed that local neurons can sense these signals directly. (3) Alternatively, signals are sensed by tanycytes and a change in tanycyte activity is sensed by neurons.

(Fig. 5). The PNS connects with the CNS from the bottom of the spinal cord to the most rostral part of the brainstem via the spinal and cranial nerves, such as cranial nerve X or vagus nerve, which is the longest nerve that connects the brain with the neck and internal organs within the chest and the peritoneal cavity (reviewed elsewhere[82]).

Thanks to expression of receptors such as TNFR1, IL-1R and Toll-like receptors[83–85], as well as its direct exposure to peripheral cells and signals, the PNS can easily detect and convey peripheral information to specific brain regions. For example, TNF, IL-1β and IL-10 were reported to activate distinct neuronal populations of the vagus nerve to inform the brain of a developing inflammatory response[86]. Furthermore, the intestinal microbiota communicates with the gut and the CNS — the 'microbiota–gut–brain axis' — to affect mental illnesses such as depression, anxiety and dementia[87–89]. In this context, immune cells could also indirectly influence brain activity by modulating what the neurons sense; for example, intestinal γδ T cells can induce repetitive or compulsive behaviour in mice by suppressing microbial metabolites that activate striatal neurons[90]. Of note, the brain can, in turn, modulate immune responses via peripheral nerves. For example, neurons in the brain that express corticotropin-releasing hormone have been shown to control plasma cell formation via acetylcholine released by the splenic nerve[91]. Other examples can be found in the colon and caecum, where the number of RORγ$^+$ T_{reg} cells was found to be reduced after TRPV1-expressing sensory neurons were activated[92], or in the meninges, where $Na_v1.8$-expressing nociceptive sensory neurons were shown to release calcitonin gene-related peptide during bacterial meningitis, which suppressed immune responses and promoted bacterial invasion[93].

This type of immune–brain communication is not systemic and relays information on the location of the stimulus, which travels rapidly owing to the electrical pulses of the PNS that reach the CNS near-instantaneously, therefore making it ideal for responses to which the body needs to act quickly and precisely. Yet, we are only starting to understand the immune contribution to physiological peripheral nerve signalling and its consequences for brain function and behaviour.

Coordination across immune–central nervous system communication pathways

To account for the need to communicate with the immune system while protecting the delicate neuronal architecture, the CNS evolved a diversity of complex anatomical routes that allow for indirect communication. Why is this diversity necessary? One reason could be the different speeds of communication through the different routes: for example, signals can be passed quickly via nerves (such as the vomiting reflex[94]), within hours via the CVOs (such as sickness behaviour[74]) or in terms of days (such as closing of the CP fenestrations in chronic immune perturbation[36]). Another reason could be the extent of the response: the activity of parenchymal immune cells may be limited to their immediate vicinity, but immune sensing by the CP, which can affect the composition of the CSF, may have a much broader effect on the CNS. Finally, cytokines may have diverse functions in the CNS depending on the context and the presence of diverse routes may be needed to ensure that cytokine activity is regulated by time, place and concentration (Box 1).

Many questions remain unanswered regarding the entry and distribution of immune cells within the CNS. The heterogeneity and proportions of different immune cell types in the CSF and brain parenchyma are different than those in the blood, which suggests active recruitment or, at least, repulsion of specific cell types. In the dura mater, immune cells are recruited directly from the skull marrow (especially monocytes, neutrophils and B cells) or from the circulation at the level of the dural sinus and are retained in the dura mater[55]. A recent study demonstrated that dural border cells produce chemorepellents for immune cells, such as semaphorin 3A (SEMA3A) and SEMA3D; local downregulation of SEMA3A expression led to increased infiltration of neutrophils in the leptomeninges of healthy mice[70]. Confocal imaging of CD4$^+$ T cells in the brain parenchyma of adult mice captured some of these cells between the basement membranes of blood vessels and leptomeninges, as well as within and beyond astrocytic endfeet, which points to their ability to access the brain from the CSF in the perivascular and subarachnoid space. Earlier studies suggested that T cells can cross the BBB endothelium in physiological conditions[3]. Furthermore, a parabiosis experiment revealed that most parenchymal T_{reg} cells die hours to days after entry, which suggests that there is continuous physiological infiltration of T cells to the brain[3].

Investigating physiological immune cell trafficking to and within the CNS remains challenging for several reasons. Each border has unique roles and properties, and it was proposed that immune cells in the CNS can acquire different phenotypes on entry, depending on which entry route they use[95]. However, the border tissues can also be viewed as being anatomically continuous; for example, the CP stroma can be considered to be an extension of the meninges, whereas the

perivascular space of the BBB opens into the subarachnoid space and was recently proposed to continue into the dura mater via ACE points (Box 2; reviewed elsewhere[96]). Furthermore, given the limited number of peripheral immune cells in the healthy CNS, the molecular mechanisms of immune cell attraction and entry are better understood in the context of CNS infection, injury or cancer[45,58,70,97]. For example, a recent live imaging study captured neutrophils and monocytes crossing the CP epithelium to enter the CSF following an intracerebroventricular injection of lipopolysaccharide to mimic meningitis[98]. One can only speculate whether the same mechanisms of entry are used at a lower scale in physiological conditions or if they are completely different.

In the next sections, we give examples of the key roles of these diverse immune–brain routes in regulating brain development and ageing, and further highlight the unanswered questions concerning the immune control of these physiological states.

Peripheral control of brain development

Peripheral signals tightly orchestrate the complex process of brain development through the same anatomical routes that later maintain adult brain physiology. The BBB, arachnoid barrier and epithelium of the CP emerge embryonically and continue their maturation during postnatal development[99–101]. This may allow peripheral signals to access the brain directly by crossing the BBB or via the CSF[101], which coincides with higher CSF protein levels in neonates[102]. These barriers have, therefore, been negatively portrayed as 'immature' or 'leaky' during development, allowing access of pathogens to the developing brain[101]. However, this permeability may be an adaptive mechanism used by immune signals to have more sustained effects on the developing brain during this period. In addition, the meningeal lymphatics, which facilitate CSF drainage, mature postnatally in all parts of the skull and around the spinal cord, through a process that was suggested to be partially driven by group 3 innate lymphoid cells (ILC3s). This delay in maturation during which CSF drainage may be 'ineffective' could also lead to the retention of immune-derived factors in the CNS during this developmental window[103,104].

Different brain regions require tailor-made signals to guide their development; microglia were once thought to be the sole immune mediators of this process through synaptic pruning and the release of cytokines and growth factors[105], yet growing evidence highlights peripheral signals as key drivers of brain development. IL-13-producing meningeal group 2 ILCs (ILC2s) were shown to be highly abundant during early postnatal development, and mice lacking ILC2s or IL-13R expression on inhibitory neurons showed defective synapse maturation in the somatosensory cortex and reduced sociability, which suggests a role for ILC2s in shaping cortical circuitry early in life[106]. Mast cells were demonstrated to infiltrate the preoptic area and respond to oestradiol, a hormone important for masculinization. In this study using neonatal male rats, oestradiol signalling in mast cells induced their degranulation and release of histamine, which caused microglia to signal to neurons in the preoptic area to promote masculinization and determine adult male sexual behaviour[107]. Naive conventional T cells and T_{reg} cells were shown to infiltrate the brain parenchyma in embryonic and neonatal stages and acquire a tissue-resident memory T cell phenotype. Antibody-mediated depletion of $CD4^+$ T cells at day 5 after birth (but not at 3 weeks) resulted in microglia maintaining a fetal phenotype, which suggests a role for T cells in instructing microglia maturation[3]. Furthermore, B cells are highly abundant at birth in the CP, meninges, parenchyma and spinal cord and were proposed to promote oligodendrocyte precursor cell proliferation through IgM–FcR signalling[108]. This further highlights that the permissive nature of border tissues is a feature, not a constraint, of the developing brain. Peripheral immune signals are therefore arranged both in space and time to operate within distinct regions and at different stages of development to exert specific and essential functions for normal CNS formation.

Beyond endogenous signals, maternal signals, transferred through the placenta or breastmilk, add another layer of regulation to brain development. Maternal lymphocytes were shown to infiltrate the hippocampus of their offspring and interact with microglia to promote social communication and cognitive function development[109]. Both mouse and human breastmilk contains many immune mediators, including immunoglobulins and complement, as well as macrophages[110]. Given their roles in the CNS, one can speculate that maternal complement and macrophages may help to prune peripheral nerves of the enteric nervous system[111], whereas antibodies may contribute to Schwann cell maturation and axon myelination of peripheral nerves, thus creating a unique immune niche that may be supportive for neurodevelopment. The full extent to which maternal cues are required to promote CNS development is unknown. Considering the permeability of CNS barriers at this age, and the presence of cytokine transporters at the BBB, signals may access the brain directly through these routes or affect CNS development indirectly by influencing the PNS or the CNS borders. Consequently, the vertical transfer of immune cells and signals from parent to offspring can confer protection against pathogens during this vulnerable period[112] while simultaneously exerting homeostatic functions that are essential for proper CNS formation.

Peripheral contribution to brain ageing

As we describe in earlier sections of this Review, various physiological mechanisms of brain development and function are regulated by immune pathways. In ageing, inflammatory mediators in the CNS and

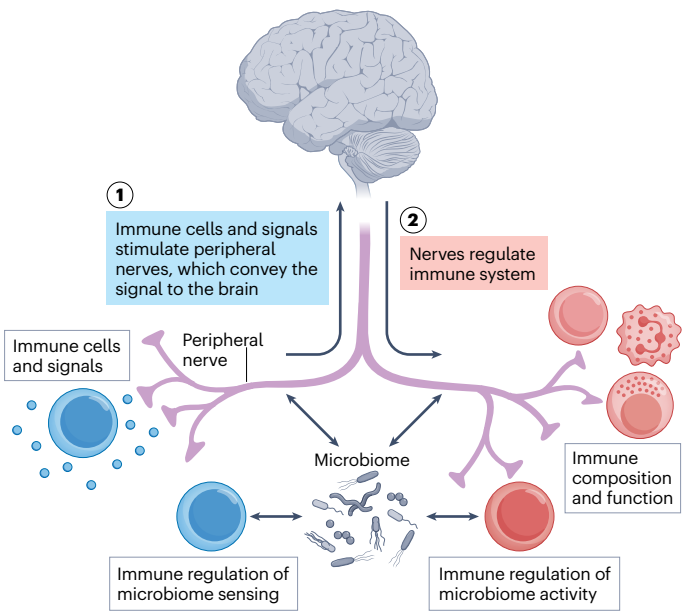


Fig. 5 | Immune–brain communication along peripheral nerves. (1) Peripheral nerves sense immune cues and transmit the information on immune status to distinct brain areas near-instantaneously. (2) In turn, the brain can regulate immune responses. In addition, the immune system can shape the ability of nerves to sense other stimuli, such as microbial products.

Table 1 | Comparison of immune–brain communication pathways in development and ageing

Development	Ageing
Barrier tissues	
'Open' barrier tissues undergo significant remodelling and maturation during development, which potentially allows for access of immune signals to the brain[99–101,103]	'Leaky' blood–brain barrier (BBB), impaired lymphatic drainage and increased infiltration of immune cells and signals into the brain parenchyma[120,122,129,134]
BBB	
Regulated transport across the barrier? Immune cells and signals promote BBB development?	Uncontrolled transport across the BBB[121] Increased leukocyte infiltration through the BBB? Impaired perivascular macrophage activity perturbs cerebrospinal fluid flow[25]
Meninges	
Meningeal IL-13 shapes synapse maturation and pruning[106] Meningeal group 3 innate lymphoid cells promote dural lymphatic development in the perinatal period[104] B-1a cells in the meninges, choroid plexus and parenchyma promote oligodendrogenesis through IgM signalling[108]	Changed immune landscape contributes to synapse loss? Increased meningeal IFNγ expression drives ageing-associated meningeal lymphatic impairment[129] Loss of CCR7 drives lymphatic decline[130] Infiltration of central nervous system (CNS)-reactive peripheral ageing-associated B cells in the dura mater[54]
Skull marrow	
Contribution of skull bone marrow-derived signals to brain development? Early progenitor B cells are more abundant in the bone marrow and meninges of 3-week-old mice compared with their 3-month-old adult counterparts[56]	Contribution of skull bone marrow-derived signals to brain ageing? Skull marrow haematopoietic cell output changes in ageing and neurodegeneration[45]
Circumventricular organs (CVOs)	
Regulation of CVO development, maturation and permeability by immune signals?	Inflammageing drives CVO malfunction in ageing?
Choroid plexus (CP)	
Immune signals promote CP maturation and function?	Type 2 immune signals dysregulate CP function and promote ageing[41] CP-derived type I IFNs promote cognitive decline via microglia[126,127]
Parenchyma	
CCL5 and IFNγ direct neuronal connectivity and social behaviour? Parenchymal T cells drive microglial maturation[3] Parenchymal mast cells shape neuronal circuitry[107] Type I IFN signalling in microglia promotes cortical development[135]	CCR5 ligands and IFNγ promote synapse and memory loss? Increased T cell infiltration in the aged brain inhibits neurogenesis through IFNγ[134] Mast cell depletion ameliorates symptoms in a mouse model of Alzheimer's disease[136]
Peripheral nervous system (PNS)	
Macrophages prune peripheral nerves of the enteric nervous system[111] Immune signals shape PNS development through similar processes as in the CNS?	Ageing of peripheral immunity drives ageing of the PNS? Dysfunctional relay of signals from the periphery to the CNS?

a dysregulated immune system in the periphery hijack these pathways, which leads to a gradual loss of brain function. The notion that the ageing of the body drives brain ageing comes from experiments showing that systemic interventions, such as heterochronic parabiosis, administration of plasma or bone marrow transplant from young animals, can restore brain function in their ageing counterparts[8,9,113–116]. The re-establishment of cognitive function in old mice following these manipulations was attributed to increased neurogenesis[8,113] and synaptic plasticity[9], improved brain vascularization[113,115] and myelination[117]. Some of these effects were mediated by changes in systemic concentrations of cytokines and chemokines; for example, plasma levels of CCL11 correlated with reduced neurogenesis in the hippocampus across ages[8]. Circulating levels of CCL11 in old mice were reduced following bone marrow transplantation from young donors[116], whereas administration of CCL11 in young mice was reported to induce cognitive impairment[8]. CCL11 is known to cross the BBB[8,9], promote microglia activation[116] and is an agonist of CCR5 (ref. 118). As mentioned earlier, CCR5 signalling on neurons is a constitutively active mechanism of memory regulation, yet in ageing, uncontrolled CCR5 activation may lead to memory loss[4]. Another source of chemokines that destabilize CCR5 signalling may be $CD8^+$ T cells, which are increasingly present in the brain with ageing and contribute to the loss of brain function via activation of the microglial synaptic pruning mechanism[119], while impairing the microglial ability to maintain myelin[120].

In homeostasis, the prevalent mechanisms of transport across the BBB are highly specific and receptor-dependent, whereas in ageing there is a shift towards nonspecific transcytosis[121], which allows access of molecules that may dysregulate brain function while restricting the access of others[122–124]. Heterochronic parabiosis experiments have shown that young blood stimulates vasculature remodelling and increases cerebral blood, effects that are mimicked by systemic treatment with the 'rejuvenation factor' growth differentiation factor 15 (GDF15)[113]. In another study, plasma transfer from old to young animals induced cognitive loss and vascular expression of VCAM1, a molecule that mediates immune cell adhesion to blood vessels, whereas blockade

of VCAM1 reversed ageing-related memory loss[115]. Analysis of the CP immune composition in ageing mice revealed an increase in ILC2s and T helper 2 cells and increased levels of IL-4 and CCL11, which were associated with epithelial inflammation and loss of barrier integrity[41,125]. By contrast, a transcriptomics study revealed increased activity of the type I IFN pathway[126]. Blockade of type I IFN receptor in the CSF of old mice restored their cognitive ability, whereas overexpression of type I IFNs specifically in the CP epithelium induced premature synaptic and cognitive loss in young mice in a microglia-dependent manner[126,127]. Heterochronic parabiosis performed in the same study revealed that young blood stimulated chemotaxis and leukocyte migration pathways in the CP of old parabionts, yet the role of this signalling at the ageing CP was not investigated[126]. Intracerebroventricular administration of young CSF into old mice was sufficient to induce oligodendrocyte precursor cell differentiation and long-term memory consolidation in the hippocampus, with fibroblast growth factor 17 identified as a key neuron-derived factor that is involved in these effects[128]. In the meninges, ageing is associated with changes in immune composition, especially an increase in the total number of T cells and levels of IFNγ, paralleled by loss of CCR7 expression[129,130]. Local overexpression of IFNγ and haematopoietic cell-specific CCR7 deficiency in young mice led to reduced lymphatic flow, which is typically associated with ageing[129,130], but the mechanisms that link immune dysregulation and structural lymphatic dysfunction in the meninges are unclear.

CVOs, which are directly exposed to the changing environment in the blood, also react to ageing. Cytokine production, antigen processing and presentation, leukocyte adhesion and leukocyte migration pathways were enriched in the median eminence of 12-month-old mice compared with 3-month-old mice in a transcriptomic analysis. Also, signalling induced by IL-1α, IL-1β, oncostatin M and TNF was specifically upregulated in the microglia of those mice[75]. It is unclear whether microinflammation at the CVOs can have a marked contribution to CNS ageing directly or indirectly via promoting systemic ageing through dysregulation of normal CVO function. For example, it was proposed that the accelerated brain ageing that is observed in obesity could be due to inflammation induced by the accumulation of obesity-associated macrophages in the CVOs[131,132].

Conclusion

In summary, the pathways of immune-to-brain communication seem to be open during development, rely on selective and more indirect routes throughout adulthood, and become leaky in ageing, when inflammatory mediators hijack virtually all communication pathways (Table 1). For example, $CD4^+$ T cells can infiltrate the brain in postnatal development to direct microglial maturation[3], whereas in ageing, parenchymal $CD8^+$ T cells interact with microglia to drive white matter pathology[120,133]. In accordance with this model, inflammation can re-engage the developmental interaction routes and accelerate the progression from physiology to ageing; viral infections associated with premature $CD8^+$ T cell infiltration to the brain were proposed to advance brain ageing and promote neurodegenerative disease[119].

Applying the proposed framework reveals many questions about the immune–brain communication pathways (Table 1). We know little about how immune signals shape the development of the border structures, such as meninges or the CVOs, and how ageing-related peripheral inflammation can change their physiological roles. Does the altered immune landscape of the meninges contribute to behavioural changes in ageing? How is the crosstalk between the skull and the brain established in development and challenged by ageing? Is there any contribution of maternal cells or signals to the maturation of brain border structures? Finally, how do peripheral inflammation, metabolic imbalance, systemic disease or other factors that promote brain ageing affect the brain through these communication routes?

Traditionally neurological diseases are treated by drugs that can cross the BBB, despite there being so many other entry points to the CNS. A better understanding of how these routes function in physiological conditions and how they communicate with one another to ensure brain health would perhaps allow us to design more effective and more precise therapies in the future.

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